unit - III

27/02/26

Linear Transformation.

Linear Transformation: function (mapping) T from vector space V to vector space (W).

$(T: V \to W)$ is said to be a linear transformation if $T(u+v) = T(u) + T(w)$ & (preserves addition & scalar multiplication)

$T(cu) = cT(u)$

for all $u, v \in V$, $c \in \mathbb{R}$

Ex:- Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$T(x, y, z) = (x+y, 2z)$.

Let $u, v \in \mathbb{R}^3$

$\Rightarrow u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2)$

$u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$

$T(u+v) = (x_1+x_2+y_1+y_2, 2(z_1+z_2)) \to (1)$

$T(u) = (x_1+y_1, 2z_1)$

$T(v) = (x_2+y_2, 2z_2)$

$T(u)+T(v) = (x_1+y_1+x_2+y_2, 2(z_1+z_2))$ (2)

from (1) & (2)

$T(u+v) = T(u) + T(v)$.

~~Let $c \in \mathbb{R}$ $cu = c(x_1+y_1, 2z_1) = (cx_1+cy_1, 2cz_1)$~~

~~$T(cu) = c(x_1$~~

Let $c \in \mathbb{R}$, $cu = c(x_1, y_1, z_1) = (cx_1, cy_1, cz_1)$

$T(cu) = T(cx_1, cy_1, cz_1)$
$= (cx_1 + cy_1, 2cz_1) \rightarrow (3)$

$cT(u) = c(x_1 + y_1, 2z_1)$
$= (cx_1 + cy_1, 2cz_1) \rightarrow (4)$

from (3) & (4)
$T(cu) = cT(u)$

$\therefore$ T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

Note:- A function $T: V \rightarrow W$ is a linear transformation if and only if $T(cu+v) = cT(u) + T(v)$ for all $c \in \mathbb{R}$ & $u, v \in V$

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2) $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by
$T(x, y) = xy$

$\Rightarrow$ Let $u, v \in \mathbb{R}^2$
$\Rightarrow u = (x_1, y_1)$ $v = (x_2, y_2)$ ~~(Let $c \in \mathbb{R}$)~~
~~$(T(cu+v) = cu = c(x_1 y_1))$~~

$u+v = (x_1+x_2, y_1+y_2)$

$T(u+v) = (x_1+x_2)(y_1+y_2) \rightarrow (1)$

$T(u) = x_1 y_1$

$T(v) = x_2 y_2$

$T(u) + T(v) = x_1 y_1 + x_2 y_2 \rightarrow (2)$

from (1) & (2)

$T(u+v) \neq T(u) + T(v)$.

$\therefore$ T is not a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}$.

<u>Note</u>:- If T is a linear transformation from V to W then,

$T(0) = 0$

3) A function $T: \mathbb{R}^3$ to $\mathbb{R}^2$ defined as

$T(x, y, z) = (x+1, y+z)$

a) $T(0,0,0) = (0+1, 0+0)$

$= (1,0) \neq (0,0)$

$\therefore$ T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

① check whether the functions are linear transformations or not.

① $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by
$T(x_1, x_2, x_3) = (2x_1 - 3x_2,\ x_1 + 4,\ 5x_3)$

$\Rightarrow$ $T(0,0,0) = (2(0) - 3(0),\ 0+4,\ 5(0))$
$= \underline{\underline{(0, 4, 0)}} \neq (0,0,0)$

$\therefore$ T is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

② $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by
$T(x, y, z) = (x+y,\ y+z)$ $\quad T(cu+v) = cT(u)+T(v)$

$\Rightarrow$ Let $u, v \in \mathbb{R}^3$ $\Rightarrow$ $u = (x_1, y_1, z_1)$
$v = (x_2, y_2, z_2)$

$cu = c(x_1, y_1, z_1) = (cx_1, cy_1, cz_1)$
$cu + v = (cx_1, cy_1, cz_1) + (x_2, y_2, z_2)$
$= (cx_1 + x_2,\ cy_1 + y_2,\ cz_1 + z_2)$

$T(cu+v) = (cx_1 + x_2 + cy_1 + y_2,\ cy_1 + y_2 + cz_1 + z_2)$

$cT(u) = c(x_1 + y_1,\ y_1 + z_1) = (cx_1 + cy_1,\ cy_1 + cz_1)$

$cT(u) + T(v) = (cx_1, cy_1, cz_1) + (x_2, y_2, z_2)$

$= (cx_1 + x_2, cy_1 + y_2, cz_1 + z_2) \rightarrow (2)$

from (1) & (2)

$T(cu+v) = cT(u) + T(v)$

$cT(u) + T(v) = (cx_1 + cy_1, cy_1 + cz_1)\ (x_2 + y_2, y_2 + z_2)$

$= (cx_1 + cy_1 + x_2 + y_2, cy_1 + cz_1 + y_2 + z_2)$

$\Downarrow$ (2)

from (1) & (2)

$T(cu+v) = cT(u) + T(v)$

$\therefore$ T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

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(3) $T: V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$ defined as

$\mathbb{R}^3$ $\mathbb{R}^2$

$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 + x_3)$.

(1) $T(cu+v) = cT(u) + T(v)$

$\Rightarrow$ Let $u, v \in \mathbb{R}^3 \Rightarrow u = (x_1, x_2, x_3)$

$v = (y_1, y_2, y_3)$

$cu = c(x_1, x_2, x_3) = (cx_1, cx_2, cx_3)$

$cu + v = (cx_1, cx_2, cx_3) + (y_1, y_2, y_3)$

$= (cx_1 + y_1, cx_2 + y_2, cx_3 + y_3)$

$T(cu+v) = (cx_1 + y_1 - cx_2 - y_2, cx_1 + y_1 + cx_3 + y_3) \rightarrow (1)$

$cT(u) = c(x_1 - x_2, x_1 + x_3)$

$= (cx_1 - cx_2, cx_1 + cx_3)$

$cT(u) + T(v) = (cx_1 - cx_2, cx_1 + cx_3)(y_1 - y_2, y_1 + y_3)$

$= (cx_1 - cx_2 + y_1 - y_2, cx_1 + cx_3 + y_1 + y_3)$ ②

from ① & ②

$T(cu+v) = cT(u) + T(v)$

$\therefore$ T is a linear transformation from $V_3(\mathbb{R}) \to V_2(\mathbb{R})$.

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④ $T: \mathbb{R}^2 \to \mathbb{R}$ defined by $T(x,y) = (x+y, x)$

$\Rightarrow$ let $u, v \in \mathbb{R}^2 \Rightarrow u = (x_1, y_1)$

$v = (x_2, y_2)$

$T(cu+v)$ $cu = (cx_1, cy_1)$

$cu + v = (cx_1, cy_1)(x_2, y_2)$

$= (cx_1 + x_2, cy_1 + y_2)$

$T(cu+v) = (cx_1 + x_2 + cy_1 + y_2, cx_1 + x_2)$ ①

$cT(u) = c(x_1 + y_1, x_1)$

$(cx_1 + cy_1, cx_1)$

$cT(u) + T(v) = (cx_1 + cy_1, cx_1)(x_2 + y_2, x_2)$

$= (cx_1 + cy_1 + x_2 + y_2, cx_1 + x_2)$ ②

from ① & ② $T(u+v) = cT(u) + T(v)$.

$\therefore$ T is a linear transformations from $\mathbb{R}^2$ to $\mathbb{R}$.

Range space & null space of linear Transformation.

Let $T: V \to W$ will be a linear transformation then the range of function T is defined by range(T). It is defined as image space of T.

i.e., $\text{range}(T) = \{T(u) \mid u \in V\}$

The null space or kernel of T is the set of vectors in V which are mapping to zero vector in W, it is denoted by ~~(ker)~~ nul(T).

i.e, $\text{nul}(T) = \{u \in V \mid T(u) = 0\}$

NOTE:- If $T: V \to W$ is a linear transformation then nul(T) is a subspace of V and range(T) is a subspace W.

ex:- $T: \mathbb{R}^2 \to \mathbb{R}^2$ defined as $T(x,y) = (x+y, 0)$ be a ~~(subspace of V)~~ linear transformation.

$\Rightarrow$ $\text{Range}(T) = \{T(u) \mid u \in \mathbb{R}^2\}$.

$= \{ T(x,y) \mid (x,y) \in \mathbb{R}^2 \}$

Range (T) $= \{ (x+y, 0) \mid x, y \in \mathbb{R} \}$

$\{ (x+y)(1,0) \mid x, y \in \mathbb{R} \}$

$= \{ k(1,0) \mid k \in \mathbb{R} \}$

Range (T) = span $\{(1,0)\}$

Basis of range (T) $= \{(1,0)\}$

dim (range (T)) = 1

Nul space (T) $= \{ u \in \mathbb{R}^2 \mid T(u) = 0 \}$

$\{ (x,y) \in \mathbb{R}^2 \mid T(x,y) = (0,0) \}$

$\{ (x,y) \in \mathbb{R}^2 \mid (x+y, 0) = (0,0)$

$x + y = 0$

$x = -y$ or

$y = -x$

$\{ (x,y) \mid y = -x, \ x, y \in \mathbb{R} \}$

$\{ (x, -x) \mid x \in \mathbb{R} \}$

$T(2,-2) = 2-2$

$= 0$

$\{x(1,-1) \mid x \in \mathbb{R}\}$

span$\{(1,-1)\}$

Basis of nul(T) is $\{(1,-1)\}$

$\underline{\dim(\text{nul}(T)) = 1}$

① Let a linear Transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined from $T(x,y,z) = (y-x, y-z)$ find the basis & the dimension of nullspace & range space (T).

⇒ matrix linear transformation T with respect to std. basis is,

Std. basis of $\mathbb{R}^3$ is $\{(1,0,0), (0,1,0), (0,0,1)\}$

$T(1,0,0) = (-1,0)$

$T(0,1,0) = (1,1)$

$T(0,0,1) = (0,-1)$

$$A = \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \begin{bmatrix} \boxed{-1} & 1 & 0 \\ 0 & \boxed{1} & -1 \end{bmatrix} \end{array}$$

Range (T) = col(A)

Nul(T) = nul(A)

Range (T) = col(A)

= span$\{(-1,0), (1,1), (0,-1)\}$

Basis of range space of T = Basis of col Pivot colⁿ

$= \{(-1, 0), (1, 1)\}$

$\therefore$ dim (Range (T)) = 2

Null (T) = Null (A) = solⁿ of set of $Ax = 0$

To find x, solve $Ax = 0$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-x + y = 0$

$y - z = 0$

$y - x = 0 \Rightarrow \underline{y = x}$

$\underline{\underline{y = z}}$

$\therefore \underline{\underline{x = y}} = z = k$ (free variable)

$X = (x, y, z) = (k, k, k)$

$Nul(T) = \{(k, k, k) \mid k \in \mathbb{R}\}$.

$\{k(1,1,1) \mid k \in \mathbb{R}\}$

$\therefore \text{Span}\{(1,1,1)\}$

Rank - Nullity theorem.

Let $T: V \to W$ be a linear transformation, then $\dim(V) = \text{Rank}(T) + \text{Nullity}(T)$

Q) Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be a L.T defined by $T(x,y,z) = (x+y+z,\ x+2y+3z)$ then find rank-nullity of T.

$\Rightarrow$ Rank of the L.T,
matrix of T w.r.t std. basis $\{(1,0,0),(0,1,0),(0,0,1)\}$ & $\{(1,0),(0,1)\}$ is $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

$\text{Range}(T) = \text{col}(A) = \text{span}\{(1,1),(1,2),(1,3)\}$

$R_2 \to R_2 - R_1$

~~Ans~~

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Pivot cols of A are (1,1) (1,2)

$\therefore$ Basis of col(A) = {(1,1) (1,2)} = Basis of range(T).

$\therefore$ Rank (T) = dim (range(T)) = 2

from rank + nullity theorem,

$$\dim(\mathbb{R}^3) = \text{Rank}(T) + \text{Nullity}(T)$$

$$3 = 2 + \text{Nulity}(T)$$